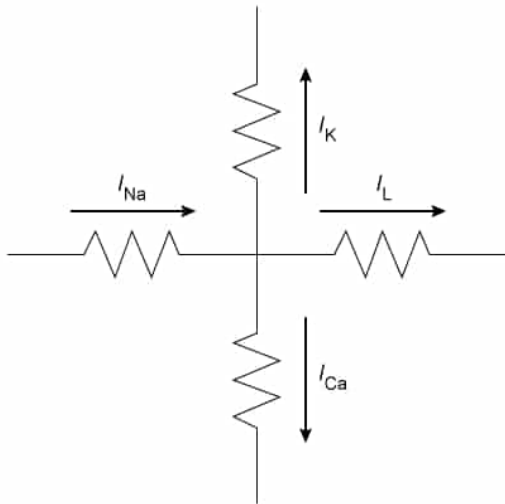


A portion of the Hodgkin-Huxley circuit model of a neuron is shown below, where I_{Na} , I_K , I_{Ca} , and I_L represent the currents through different cell membrane channels.



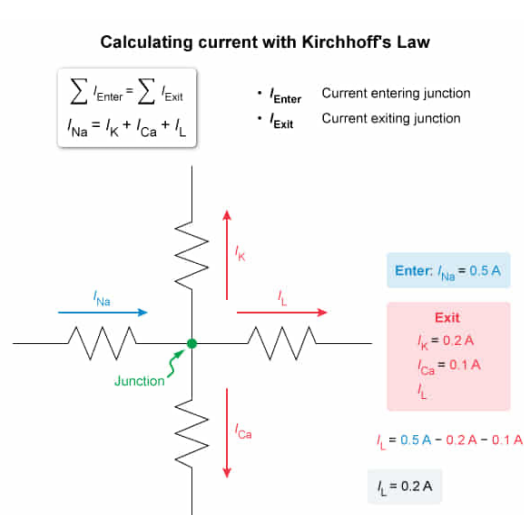
What is the value of I_L when $I_{Na} = 0.5$ A, $I_K = 0.2$ A, and $I_{Ca} = 0.1$ A?

- ☐ A. 0.8 A [5%]
- ☐ B. 0.3 A [10%]
- ✓ ☒ C. 0.2 A [81%]
- ☐ D. 0.1 A [2%]

Correct

82% answered correctly

Explanation:



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Junctions are points in a circuit where three or more circuit elements are connected. According to [Kirchhoff's law](#), the sum of all [currents](#) entering a junction I_{Enter} must equal the sum of all currents exiting that junction I_{Exit} :

$$\sum I_{\text{Enter}} = \sum I_{\text{Exit}}$$

Kirchhoff's law results from the conservation of **electric charge** at a circuit junction, which states that charge cannot be created or destroyed at circuit junctions. Because current represents the flow of charges, current flowing into a junction must be balanced by an equal amount of current flowing out of the junction.

In this question, the circuit elements are all connected at the center of the diagram and form a junction. The direction of the arrow indicates the direction of current flow. The arrow for I_{Na} points toward the junction, so this current enters the junction. In contrast, the arrows for I_{K} , I_{Ca} , and I_{L} point away from the junction, so these currents exit the junction. Hence, Kirchhoff's law requires:

$$I_{\text{Na}} = I_{\text{K}} + I_{\text{Ca}} + I_{\text{L}}$$

Solving this equation for I_{L} yields:

$$I_{\text{L}} = I_{\text{Na}} - I_{\text{K}} - I_{\text{Ca}}$$

Substituting the values for the currents into this equation gives:

$$I_{\text{L}} = 0.5 \text{ A} - 0.2 \text{ A} - 0.1 \text{ A} = 0.2 \text{ A}$$

Therefore, I_{L} equals 0.2 A.

(Choice A) 0.8 A is calculated by summing all the current values given. However, I_{K} and I_{Ca} are exiting the junction and should be subtracted from I_{Na} instead of added to it.

(Choice B) 0.3 A results from adding I_{K} and I_{Ca} , but this value is subtracted from I_{Na} to calculate I_{L} .

(Choice D) 0.1 A is obtained by subtracting I_{Ca} from I_{K} instead of subtracting both these values from I_{Na} .

Educational objective:

Circuit junctions are points where three or more elements meet. By Kirchhoff's law, the sum of the currents entering a junction equals the sum of the currents exiting the junction.

Time spent: 0 sec
Subject:
Physics

QID: 403816
System:
Electrostatics and Circuits